

Paper 2 - Set 04 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	D	Circles and circle theorems
2	E	Indices and surds
3	A	Functions and transformations
4	H	Straight lines
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7	A	Circles and circle theorems
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14	F	Indices and surds
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20	H	Chains of reasoning

Worked solutions

Question 1 CIR-04 · Circles and circle theorems · spec MM3.2

The points $(1, 7)$ and $(5, -1)$ are the ends of a diameter of a circle. Find the equation of the circle.

Answer: D

Centre is the midpoint $(3, 3)$; $\text{radius}^2 = (3 - 1)^2 + (3 - 7)^2 = 4 + 16 = 20$.

Question 2 IND-10 · Indices and surds · spec MM1.1

The value of $\frac{2^{n+2} + 2^n}{2^{n+1}}$ is

Answer: E

Factor 2^n : numerator $= 2^n(4 + 1) = 5 \cdot 2^n$; denominator $= 2 \cdot 2^n$. Quotient $= \frac{5}{2}$.

Question 3 FUN-02 · Functions and transformations · spec MM8.2

$f(x) = x^2 - 4x$. The graph of $y = f(x + 2)$ has its minimum point at

Answer: A

$f(x) = (x - 2)^2 - 4$ has minimum at $(2, -4)$. Replacing x by $x + 2$ translates the graph 2 units to the left: $(0, -4)$.

Question 4 X2-05 · Straight lines · spec MM3.1

Two distinct lines have equations $y = m_1x + c_1$ and $y = m_2x + c_2$. Which of the following statements is/are true?

- I If $m_1m_2 = -1$ then the lines meet at exactly one point.
- II If $m_1 = m_2$ then the lines do not meet.
- III If the lines meet on the y -axis then $c_1 = c_2$.

Answer: H

I: perpendicular lines are not parallel, so they meet once. II: equal gradients and distinct lines means parallel and different, so no intersection. III: on the y -axis $x = 0$, so $y = c_1 = c_2$.

Question 5 X2-17 · Integration · spec MM7.5

The trapezium rule with n strips is used to estimate $\int_0^1 \sqrt{x} \, dx$. Which of the following is true for every n ?

Answer: B

$y = \sqrt{x}$ is concave (bends downwards), so every chord lies below the curve and the trapezia underestimate the area.

Question 6 X2-25 · Identifying errors in proofs · spec Err2

A student solves $x^2 - 5x + 6 > 0$ as follows.

I $x^2 - 5x + 6 = (x - 2)(x - 3)$.

II So $(x - 2)(x - 3) > 0$.

III So $x - 2 > 0$ and $x - 3 > 0$.

IV So $x > 3$.

Which of the following is correct?

Answer: A

The full solution set is $x < 2$ or $x > 3$.

Question 7 X2-10 · Circles and circle theorems · spec MM3.2

The circle C has centre (a, b) and radius r . P : ' $r > |b|$ '. Q : ' C crosses the x -axis at two distinct points'. Which one of the following is true?

Answer: A

The distance from the centre to the x -axis is $|b|$; the circle cuts the axis twice exactly when this is less than r .

Question 8 PRF-17 · Mathematical proof · spec Prf1

Which of the following claims is FALSE, and therefore can be disproved by a counterexample?

Answer: B

$a = 1, b = -2$: $1 > -2$ but $1 < 4$. The others are true.

Question 9 LGC-17 · The logic of arguments · spec Arg1

Let x be a real number. Which of the following statements is equivalent to $x \geq 2$ if and only if $x^3 \geq 8$?

Answer: B

'If and only if' is the conjunction of the implication and its converse.

Question 10 LGC-24 · The logic of arguments · spec Arg4

Let S be the statement: 'every even number greater than 2 can be written as the sum of two primes'. A single counterexample to S would be

Answer: E

To refute 'for all n , $P(n)$ ' you exhibit one n with $P(n)$ false.

Question 11 RSN-20 · Chains of reasoning · spec Prf5

n is a positive integer such that n^2 has exactly 3 positive divisors. Which of the following must be true?

Answer: B

A number with exactly 3 divisors is the square of a prime; so $n^2 = p^2$ and $n = p$ is prime.

Question 12 ERR-15 · Identifying errors in proofs · spec Err1

Claim: the sum of the first n odd numbers is n^2 .

- I The first odd number is $1 = 1^2$, so the claim holds for $n = 1$.
- II Suppose the claim holds for some n , i.e. $1 + 3 + \dots + (2n - 1) = n^2$.
- III Adding the next odd number, $2n + 1$, gives $n^2 + 2n + 1 = (n + 1)^2$.
- IV So the claim holds for $n + 1$, and hence for all positive integers n .

Which of the following is correct?

Answer: A

This is a valid proof by induction: a base case, and a step showing that truth for n implies truth for $n + 1$.

Question 13 DIF-11 · Differentiation · spec MM6.3

Which of the following statements about $f(x) = x^3 - 3x$ is/are true?

- I f is increasing for all $x > 1$.
- II f has a local maximum at $x = -1$.
- III $f(x) \geq -2$ for all real x .

Answer: E

$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$: positive for $x > 1$ (I true); $f'' = 6x < 0$ at $x = -1$ (II true). III false: the local minimum $f(1) = -2$ is not a global minimum since $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ (e.g. $f(-3) = -18$).

Question 14 IND-16 · Indices and surds · spec MM1.2

Which of the following statements is/are true for all positive real numbers a and b ?

- I $\sqrt{a}\sqrt{b} = \sqrt{ab}$
- II $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$
- III $\sqrt{a^2 + b^2} < a + b$

Answer: F

I is a law of surds. II fails for $a = b = 1$: $\sqrt{2} \neq 2$. III: $(a + b)^2 = a^2 + b^2 + 2ab > a^2 + b^2$ since $ab > 0$, and both sides are positive, so III is true.

Question 15 DIF-19 · Differentiation · spec MM6.1

Which of the following statements is/are true for every function f that has a derivative at every real number?

- I If $f'(a) = 0$ then f has a maximum or a minimum at $x = a$.
- II If f has a local minimum at $x = a$ then $f'(a) = 0$.
- III If $f'(x) > 0$ for all x , then $f(3) > f(2)$.

Answer: G

I is false: $f(x) = x^3$ has $f'(0) = 0$ but no turning point. II is true (a stationary point is required at an interior local extremum of a differentiable function). III is true: a strictly increasing function has $f(3) > f(2)$.

Question 16 RSN-15 · Chains of reasoning · spec Prf5

A sequence of 10 integers is such that the sum of any three consecutive terms is positive and the sum of any five consecutive terms is negative. Which of the following statements must be true?

- I The sum of all 10 terms is negative.
- II The sequence contains at least one negative term.
- III The sequence contains at least one positive term.

Answer: H

Terms 1-5 and 6-10 are two blocks of five, each with negative sum (I). A five-block has negative sum, so some term is negative (II). A three-block has positive sum, so some term is positive (III).

Question 17 ERR-08 · Identifying errors in proofs · spec Err1

Claim: if n is an integer and n^2 is divisible by 4, then n is divisible by 4.

- I Suppose n^2 is divisible by 4.
- II Then $n^2 = 4k$ for some integer k .
- III So $n = 2\sqrt{k}$, and since n is an integer $\sqrt{k}$ is an integer.
- IV Hence n is even, so $n = 2m$, and $n^2 = 4m^2$ is divisible by 4, so n is divisible by 4.

Which of the following is true?

Answer: A

$n = 2$: $n^2 = 4$ is divisible by 4 but n is not. Lines I-III are correct deductions (n even). Line IV concludes ' n divisible by 4' from ' n^2 divisible by 4', which is exactly the unproved claim - circular and false.

Question 18 POL-08 · Polynomials, factor and remainder theorems · spec MM1.6

Which of the following statements about the polynomial $p(x) = x^3 - 6x^2 + 11x - 6$ is/are true?

- I $p(x)$ has three distinct real roots.
- II $(x - 4)$ is a factor of $p(x) - 6$.
- III $p(x) > 0$ for all $x > 3$.

Answer: H

$p(x) = (x - 1)(x - 2)(x - 3)$, so I is true. $p(4) - 6 = 3 \cdot 2 \cdot 1 - 6 = 0$, so II is true. For $x > 3$ all three factors are positive, so III is true.

Question 19 LGC-12 · The logic of arguments · spec Arg3

Consider the statement: **there exists a real number k such that for all real x , $x^2 + kx + 1 > 0$.** Which of the following is true?

Answer: A

Need discriminant $k^2 - 4 < 0$, i.e. $-2 < k < 2$. So some (not all) values of k work; $k = 0$ gives $x^2 + 1 > 0$. ($k = 2$ gives $(x + 1)^2$, which is 0 at $x = -1$.)

Question 20 RSN-02 · Chains of reasoning · spec Prf5

The integers a, b, c satisfy $a + b + c = 0$ and $abc = 12$. Which of the following statements must be true?

- I Exactly one of a, b, c is positive.
- II $a^2 + b^2 + c^2$ is even.
- III One of a, b, c is divisible by 3.

Answer: H

$abc > 0$ means either none or exactly two of them are negative; they cannot all be positive with sum 0, so exactly two are negative and one positive (I). $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca) = -2(ab + bc + ca)$ is even (II). 3 is prime and divides abc , so it divides one of them (III). (Example: $-1, -3, 4$.)